NEGATIVE DEPENDENCE IN ACTIVATED RANDOM WALK

Abstract. We study negative association of the stationary sleeper set of finite activated random walk. We prove negative association at small and large sleep rates on each fixed undirected sink graph, and under weak internal motion on every fixed bidirected support. On complete graphs, pairwise negative correlation and a block-event inequality hold at every size and sleep rate; full negative association holds under sufficiently weak killing. The proofs use Green determinants, escape-probability regions, positive sources in stationary equations, and avalanche response inequalities. We also give exact counterexamples to stronger dependence properties and prove conditional consequences for concentration and limiting densities. The general negative-association conjecture remains open.

1. The stationary sleeper set

Let V be a finite set of nonsink vertices, and let ∂ be an absorbing sink. An active particle jumps at rate one, with substochastic internal transition matrix K; its missing row mass is the probability of jumping to ∂. Assume that the sink is accessible from every vertex, so $\rho(K) < 1$. A lone active particle at v sleeps at rate $\lambda_v > 0$. An arrival wakes a sleeper. A stable configuration is a subset $X \subseteq V$, or its vector of indicators $(X_v)_{v\in V}$. We write $\mu_{K,\lambda}$ for the stationary law obtained by repeatedly adding an active particle and stabilizing, with uniform additions on V.

The main setting is simple random walk on a finite undirected graph, with a common sleep rate λ. General kernels and unequal rates will be allowed only where stated. Put $N_A = \sum_{v\in A} X_v$ and $N = N_V$.

Definition 1.1. A law μ on $\{0,1\}^V$ is *negatively associated* (NA) if

$$\operatorname{Cov}_\mu(f(X_A), g(X_B)) \le 0 \tag{1.1}$$

for disjoint $A, B \subseteq V$ and coordinatewise increasing real functions f, g. It is enough to test indicators of increasing events: every increasing function on a finite cube is a constant plus a nonnegative linear combination of such indicators.

Conjecture 1.2. For every finite undirected simple graph with an accessible sink and every common $\lambda > 0$, the stationary sleeper law is negatively associated.

This asks for more than repulsion between pairs of sites. For example, it would make the events $\{N_A \ge r\}$ and $\{N_B \ge s\}$ negatively correlated for every pair of disjoint regions and every pair of thresholds. It would also give the exponential-moment and concentration bounds proved in Section 8. These consequences concern fluctuations around the mean; identification of the mean density is a separate question.

The article gives readable proofs of the principal mechanisms. A proof that uses a longer calculation states the needed lemma explicitly and links its verification in the companion files. Section 11 identifies those files and the exact claims they support. Thus a deferred calculation is an identified input to an argument, rather than an additional conjecture.

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2. Exact sampling, factorization, and an all-rate identity

Theorem 2.1 (Levine–Liang exact sampler [1])**.** *Stabilize one active particle at every vertex of V. The resulting law is $\mu_{K,\lambda}$ and is invariant under addition at each individual vertex. It is consequently stationary for every drive; when the drive can reach all vertices, this is its unique stationary law.*

Sampling mechanism. Start from one active at each vertex and any fixed vector of extra active particles. Move only surplus particles, leaving one active at every vertex, until all extras have exited. These are legal topplings and terminate almost surely. The unused site-stack suffixes are independent fresh stacks, so the remaining stabilization has the saturated law. Abelianity identifies this with adding the extras after a saturated stabilization. This proves the invariance used below; the uniqueness assertion is the cited theorem. □

Proposition 2.2 (Strong-component factorization)**.** *Let $C_1, \dots, C_m$ be the strongly connected components of the directed support of K. Let μ_i be the stationary law on C_i with every exit from C_i killed. Then*

$$\mu_{K,\lambda} = \bigotimes_{i=1}^{m} \mu_i.$$

If the support is acyclic, this reduces to $\bigotimes_{v\in V} \mathrm{Bernoulli}(\lambda_v/(1+\lambda_v))$.

Proof. Order the components so that all edges between components point forward, and stabilize them in that order. Conditional on the previously exposed components, the next component receives its saturated input plus a fixed vector of extra arrivals. Its stacks are independent and unexposed. Theorem 2.1 makes its output law independent of those arrivals; no later component can send particles back. Iterated conditioning proves the factorization. On a DAG, process individual vertices in topological order. After routing the surplus, the last particle sleeps with probability $\lambda_v/(1+\lambda_v)$ according to a fresh instruction, and never returns after a jump. These final Bernoulli decisions are independent. □

The DAG conclusion is stronger than NA: its generating polynomial is a product of linear factors with negative real roots, hence is strongly Rayleigh. Cycles are where nontrivial stationary dependence begins.

There is also an exact formula valid on every finite killed kernel, at every positive rate. Let $G = (I-K)^{-1}$, let G_A denote its principal submatrix on A, and use $\det G_\varnothing = 1$.

Theorem 2.3 (Full occupation)**.** *For $x_v = 1/\lambda_v$ and $x_A = \prod_{v\in A} x_v$,*

$$\mu_{K,\lambda}(X = V) = \left(\sum_{A\subseteq V} \frac{x_A}{\det G_A} \right)^{-1}. \tag{2.1}$$

Self-loops, nonreversible walks, and unequal sleep rates are allowed.

Proof via an exact change of basis. Let A_v be the internal-DLA addition matrix on subsets of V: a walk started at v occupies its first empty site, or exits at the sink. Set $J_v = K(v,\partial)I + \sum_w K(v,w)A_w$. The matrices A_v commute. If $\mathcal{N}$ has column B equal to the ARW stabilization law from one active at every vertex of B, define

$$Me_B = \prod_{v\in B} [A_v + x_v(A_v - J_v)]e_\varnothing.$$

The first-step sleep/jump equation, together with legal surplus routing, gives $\mathcal{N}M = I$: after intertwining with ARW addition, each factor adds a sleeper deterministically at a previously empty site. Hence $\mu = M^{-1}e_V$.

No column B of M can contain a state with more than $|B|$ particles. Its last row therefore vanishes except at $B = V$, so $\mu(V) = 1/M_{V,V}$. The coefficient of x_A in $M_{V,V}$ is obtained by

filling $V \setminus A$ and applying $\prod_{v\in A}(A_v - J_v)$ on the remaining traced walk. The needed exact IDLA identity is

$$[e_A] \prod_{v\in A} (A_v - J_v)e_\varnothing = \det(I - Q), \tag{2.2}$$

where Q is that traced kernel. A constant-term application of the MacMahon master theorem [2] proves (2.2); the full coefficient calculation and its absorbing-chain uniqueness argument are in the companion full-occupation proof. Finally $(I - Q)^{-1} = G_A$ by Schur complementation. This gives the coefficient $1/\det G_A$ and proves (2.1). □

Corollary 2.4 (An all-rate monotonicity consequence)**.** *The probability of full occupation is strictly increasing in each sleep rate. If $K \le \widetilde{K}$ entrywise and both kernels are killed, then $\mu_{K,\lambda}(V) \le \mu_{\widetilde{K},\lambda}(V)$.*

Proof. The denominator in (2.1) has strictly positive coefficients in the variables x_v. For the kernel comparison, Jacobi's identity and $-\log\det(I - K_D) = \sum_{m\ge1} \operatorname{tr}(K_D^m)/m$ give

$$\log\det G_A = \sum_{m\ge1} \frac{1}{m} \sum_{\substack{\text{rooted closed walks of length } m \\ \text{visiting } A}} \text{weight}_K(\text{walk}).$$

Every weight increases with the entries of K. Each $\det G_A$ therefore increases, and the denominator in (2.1) decreases. □

Formula (2.1) is about full occupation, not all occupation cylinders. Applying it after tracing onto A gives a trace-law probability, which need not equal the original marginal on A. This distinction is important in the complete-graph argument below.

3. Small sleep rates: a Green-determinant proof

Theorem 3.1 (Small-sleep negative association)**.** *Fix a finite killed kernel K and positive numbers α_v. With $\lambda_v = t\alpha_v$, the stationary law is NA for all sufficiently small $t > 0$. The threshold depends on K and α. No reversibility assumption is needed. If K is irreducible on V, the sleeper indicators also satisfy the BK inequality in this interval.*

The key is an asymptotic determinantal structure. The full law need not be determinantal; its first nonzero occupation coefficients are.

Lemma 3.2 (Rare-sleep expansion)**.** *For every $A \subseteq V$, with $\alpha_A = \prod_{v\in A}\alpha_v$,*

$$\mu_t(X \supseteq A) = t^{|A|}\alpha_A \det G_A + O(t^{|A|+1}). \tag{3.1}$$

The same leading coefficient holds for the exact sleeper set A.

Proof. At $t = 0$, after ineffective sleep instructions are skipped, stabilization is a finite absorbing chain. A bounded sequence of sink-directed jumps has uniformly positive probability, so its absorption time has exponential tails for configurations with at most $|V| + 1$ particles. Inserting sleep coins of probability $O(t)$ and restarting after each successful sleep shows that j successful sleeps have probability $O(t^j)$. In particular, $\mu_t(X = C) = O(t^{|C|})$.

Choose $v \in A$ and put $B = A \setminus \{v\}$. Apply stationarity under addition at v. To finish with A occupied at order $t^{|A|}$, the initial exact sleeper set must be B. Smaller sets lack enough particles; other sets of size $|A| - 1$ require at least two new successful sleeps; sets of size $|A|$ require at least one. All these alternatives contribute only at higher order.

From B, the unique first-order mechanism is for the added particle to avoid B and sleep at v before absorption. Its probability is $t\alpha_v G^B(v, v) + O(t^2)$, where G^B is the Green matrix with additional killing on B. The error follows from the finite second moment of the hitting time. Schur complementation gives

$$G^B(v, v) = G(v, v) - G(v, B)G_B^{-1}G(B, v) = \frac{\det G_A}{\det G_B}.$$

Induction from the empty set proves the exact-set expansion. Larger exact sets enter the cylinder probability only at higher order. □

Lemma 3.3 (Strict disjoint Green-minor inequality)**.** *If K is irreducible and I, J are disjoint nonempty sets, then*

$$\det G_{I\cup J} < \det G_I \det G_J. \tag{3.2}$$

Proof. Put $h(D) = \log\det(I - K_D)$. Jacobi's identity gives

$$\log \frac{\det G_I \det G_J}{\det G_{I\cup J}} = h(V \setminus I) + h(V \setminus J) - h(V \setminus (I \cup J)) - h(V).$$

Expand each h as $-\sum_{m\geq 1} \operatorname{tr}(K_D^m)/m$. A rooted closed walk contributes exactly when it visits both I and J, and then contributes its positive weight divided by its length. Irreducibility supplies at least one such walk. Absolute convergence follows from $\rho(K) < 1$. □

Proof of negative association in Theorem 3.1. Let E, F be nonconstant increasing events on disjoint coordinate sets. Let a, b be their minimum witness sizes and $\mathcal{E}_a, \mathcal{F}_b$ the corresponding families of occupied witnesses. Lemma 3.2 gives the leading coefficient of the NA gap as

$$t^{a+b} \sum_{I\in\mathcal{E}_a,\, J\in\mathcal{F}_b} \alpha_I\alpha_J\big(\det G_I \det G_J - \det G_{I\cup J}\big). \tag{3.3}$$

It is strictly positive by Lemma 3.3. There are finitely many event pairs, so a single interval works for all of them. Threshold decomposition proves the function formulation. For reducible K, apply this argument inside each strongly connected component and use Proposition 2.2; independent products preserve NA. □

The pair covariance makes the mechanism particularly explicit:

$$\operatorname{Cov}_{\mu_t}(X_i, X_j) = -t^2\alpha_i\alpha_j G(i,j)G(j,i) + O(t^3). \tag{3.4}$$

The first interaction coefficient records communication in both directions. One-way communication disappears under strong-component factorization.

Definition 3.4. For increasing events E, F, their *disjoint occurrence* $E \,\square\, F$ means that two disjoint subsets of occupied coordinates witness E and F. The BK property is $\mu(E \,\square\, F) \leq \mu(E)\mu(F)$, even when the events use overlapping coordinate sets. BK implies NA by taking events on disjoint supports.

BK part of Theorem 3.1. Write $w(I) = \alpha_I \det G_I$. The leading coefficient of $\mu(E)\mu(F)$ is $\sum_{I\in\mathcal{E}_a, J\in\mathcal{F}_b} w(I)w(J)$. The leading coefficient of $\mu(E \,\square\, F)$ sums $w(C)$ over *distinct* unions $C = I \cup J$ with $I \cap J = \varnothing$. Choose one decomposition per union. Each selected product exceeds $w(C)$ by (3.2); all unused product terms are nonnegative. If there is no such union, the disjoint event starts at higher order. Finiteness again gives one interval for all event pairs. □

4. Large sleep rates: holes, escape regions, and separators

Theorem 4.1 (Large-sleep negative association)**.** *Fix a finite undirected graph with accessible sink, allowing positive edge weights and self-loops, and fix $\alpha_v > 0$. With $\lambda_v = \alpha_v/t$, the stationary law is NA for all sufficiently small $t > 0$.*

Here the sparse variables are vacancies $Y_v = 1 - X_v$. For nonempty A and $v \in A$, define

$$e_A(v) = \mathbb{P}_v(T_\partial < T_A^+),$$

where T_A^+ is the first return to A after at least one jump. Last-exit decomposition gives $G_A e_A = \mathbf{1}$.

Lemma 4.2 (Escape regions)**.** *For every finite killed kernel,*

$$F_A(t) := \mathbb{P}(Y_v = 1 \text{ for all } v \in A) = \frac{t^{|A|}}{\alpha_A} C_A + O(t^{|A|+1}), \tag{4.1}$$

where $C_\varnothing = 1$ and C_A is the volume of

$$\mathcal{R}_A = \left\{ u \in [0,\infty)^A : \text{every } \varnothing \neq B \subseteq A \text{ has some } v \in B \text{ with } u_v < e_B(v) \right\}. \tag{4.2}$$

Consequently $C_{I\cup J} \le C_I C_J$ for disjoint I, J.

Expansion and geometric argument. At $t = 0$, lone particles sleep immediately. Every vacancy requires a rare lone jump, giving the orders in (4.1) by the same finite absorption argument used above. The leading coefficients satisfy

$$\sum_{H \subseteq A} (-1)^{|A\setminus H|} C_H \prod_{v \in A\setminus H} e_A(v) = 0. \tag{4.3}$$

The companion escape-expansion calculation derives this recurrence from the stationary harmonic-polynomial identity; adding small direct killing and taking its limit handles zero escape coordinates. This is the only ARW calculation needed for the geometric step.

For that step, note that $e_A(v) \le e_B(v)$ when $v \in B \subseteq A$. Every point of $\mathcal{R}_A$ lies in a strip $u_v < e_A(v)$. Imposing strips for a nonempty set D automatically satisfies every constraint meeting D, leaving precisely $\mathcal{R}_{A\setminus D}$ on the other coordinates. Inclusion–exclusion gives (4.3) for the volumes, proving the identification. Each region is bounded by the singleton constraints and contains a small positive cube, since some escape coordinate is positive on every nonempty subset. Restriction of coordinates gives $\mathcal{R}_{I\cup J} \subseteq \mathcal{R}_I \times \mathcal{R}_J$ and hence the stated inequality. □

For $c_i = C_{\{i\}} = 1/G(i,i)$, the pair gap factors as

$$c_i c_j - C_{\{i,j\}} = \frac{G(i,j)G(j,i)e_{\{i,j\}}(i)e_{\{i,j\}}(j)}{G(i,i)G(j,j)}. \tag{4.4}$$

If each vertex can reach the sink without visiting the other, this is strict. Choosing u_i and u_j between their pair escape probabilities and c_i, c_j, and taking the other coordinates sufficiently small, produces an open box excluded from $\mathcal{R}_{I\cup J}$ but contained in $\mathcal{R}_I \times \mathcal{R}_J$. The witness argument from the small-rate proof then proves NA whenever all cross-pair escape gaps are strict.

A cut vertex creates the genuine difficulty: every escape from i may pass through j, making (4.4) zero. The following next-order lemma resolves precisely this degeneracy.

Lemma 4.3 (Separator correction)**.** *Assume the undirected nonsink graph is connected. Say that v dominates u if every path from u to the sink passes through v. Write*

$$F_A(t) = \frac{t^{|A|} C_A}{\alpha_A}(1 + t d_A + O(t^2)).$$

For disjoint nonempty I, J, the equality $C_{I\cup J} = C_I C_J$ holds exactly when every cross-pair is comparable by domination. Whenever it holds,

$$d_I + d_J - d_{I\cup J} > 0. \tag{4.5}$$

The equality condition is preserved by taking unions on either side, provided the two sides stay disjoint.

Proof outline. An incomparable pair gives the excluded open box just described. Conversely, a vertex v dominating all other selected vertices has the only nonzero escape coordinate, so (4.3) gives $C_{\{v\}\cup W} = c_v C_W$. Domination ancestors form a chain. Peeling them proves the asserted equality characterization.

Let D_v be the vertices strictly dominated by v. The technical calculation establishes

$$d_{\{v\}} + d_W - d_{\{v\}\cup W} = \kappa(v, W) > 0 \qquad (\varnothing \neq W \subseteq D_v),$$

and, when $W, J \subseteq D_v$ are disjoint with $C_{W\cup J} = C_W C_J$, $\kappa(v, W \cup J) \geq \kappa(v, W)$. The separator calculation, formulas (14)–(17), expresses $c_v\kappa(v, W)$ as sums of hitting probabilities times nonnegative volume gaps; its first sum is strictly positive. Those formulas prove both signs, including the limit where escape probabilities vanish.

Interchange I, J if necessary, peel a dominating vertex $v \in I$, and set $I' = I \setminus \{v\}$. The correction gap becomes

$$d_{I'} + d_J - d_{I'\cup J} + \kappa(v, I' \cup J) - \kappa(v, I').$$

The last difference is nonnegative. The final peeling step, when one block is exhausted, supplies a strictly positive $\kappa(v, W)$. This proves (4.5). □

Completion of Theorem 4.1. For increasing vacancy events on disjoint supports, a nonzero leading gap is positive by the minimum-witness argument. If it vanishes, all pairs of minimum witnesses satisfy the equality condition in Lemma 4.3, and their next corrections are positive. The only potentially negative extra monomial coefficients, at the next witness size, arise from unions of two minimum witnesses. Their leading cylinder gaps vanish by the union property in that lemma. All remaining next-order contributions are nonnegative. Thus every nontrivial event gap is positive at one of its first two possible orders. There are finitely many events. Finally NA is preserved by complementing all coordinates: two decreasing functions have the same covariance as their negatives. Component factorization handles disconnected nonsink graphs. □

The small- and large-sleep theorems apply to every fixed graph. Their thresholds depend on that graph; they do not assert a single perturbative interval valid along a growing sequence of volumes.

5. Weak internal motion and the first interaction term

Keep all sleep rates fixed, put $p_i = \lambda_i/(1 + \lambda_i)$ and $q_i = 1 - p_i$, and replace the internal kernel by $K_\varepsilon = \varepsilon P$. Assume $P_{ii} = 0$ and $P_{ij} > 0$ if and only if $P_{ji} > 0$. At $\varepsilon = 0$ the stationary law is the product measure π_p. For increasing f define its product-measure influence by

$$I_i(f) = \mathbb{E}_{\pi_p}[f(X^{i,1}) - f(X^{i,0})].$$

Theorem 5.1 (Weak internal motion)**.** *For any fixed loopless bidirected support and positive site rates, $\mu_{\varepsilon P,\lambda}$ is NA for all sufficiently small $\varepsilon > 0$. If the essential supports of nonconstant increasing f, g are disjoint and have finite graph distance d, then*

$$\operatorname{Cov}(f, g) = -\varepsilon^{2d} \sum_{\substack{i\in\operatorname{supp} f,\ j\in\operatorname{supp} g \\ \operatorname{dist}(i,j)=d}} (P^d)_{ij}(P^d)_{ji} p_i q_i p_j q_j I_i(f) I_j(g) + O(\varepsilon^{2d+1}). \tag{5.1}$$

Proof outline with the technical deletion step identified. Give each directed edge its own variable. Analyticity and strong-component factorization imply that a nonzero covariance monomial must contain a strong component meeting both supports. Crossing the d distance cuts in both directions requires at least $2d$ edges. At that degree, the support consists of two oppositely directed geodesics, forming a chain of cycles.

The deletion lemma in the companion weak-interaction proof, formulas (38)–(41), preserves the endpoint marginal modulo the square of each directed-edge variable. It reduces this minimal chain to the exact two-site law. The coefficient there is $-p_i q_i p_j q_j$; independent coordinates outside the chain supply the influences. Summing the path weights gives (5.1). A nearest essential pair has positive influences and positive weights in both directions, so the leading coefficient is strictly negative. Finitely many event pairs give the claimed NA interval. □

This formula quantifies how weak interactions transmit dependence over distance. It permits nonreversible edge weights; the two-way support condition is what supplies the negative leading term.

6. Complete graphs: a trace comparison at every rate

Let $V = [n]$ and let ∂ be an absorbing sink. It is convenient to allow holding jumps and use the kernel

$$K(x,y) = \frac{1}{n+a} \quad (x,y \in V), \qquad K(x,\partial) = \frac{a}{n+a}, \qquad a > 0. \tag{6.1}$$

Particles jump at rate one and lone active particles sleep at a common rate $\lambda \in (0,\infty)$. Write

$$p = \frac{\lambda}{1+\lambda}, \qquad q = 1-p,$$

and let $\mu_n^{p,a}$ be the stationary stable law for uniform additions followed by stabilization. Thus $X_x = 1$ means a sleeper at x. Equivalently, by Levine and Liang's saturated-sampling theorem [1], X is obtained by stabilizing one active particle at every site. We use that theorem as an input; the arguments below concern the resulting stationary law. For probability laws on subsets, $\sigma \preceq_{\mathrm{st}} \tau$ means that every increasing function has smaller expectation under σ.

Theorem 6.1 (Trace sandwich, block OR, and strict pair correlation)**.** *Fix $n \geq 2$, $0 < p < 1$, and $a > 0$. Let $X \sim \mu_n^{p,a}$ and $S_n = \sum_{x\in V} X_x$. For a nonempty proper set $D \subset V$, put $m = n - |D|$, $E_D = \{\sum_{x\in D} X_x > 0\}$, and let $\tau_D = \mu_m^{p,a}$ on $V \setminus D$. Then*

$$\mathrm{Law}(X_{V\setminus D}) \preceq_{\mathrm{st}} \tau_D \preceq_{\mathrm{st}} \mathrm{Law}(X_{V\setminus D} \mid X_D = 0), \tag{6.2}$$

$$\mathrm{Law}(X_{V\setminus D} \mid E_D) \preceq_{\mathrm{st}} \tau_D \preceq_{\mathrm{st}} \mathrm{Law}(X_{V\setminus D} \mid E_D^c). \tag{6.3}$$

Consequently, for every increasing $g : \{0,1\}^{V\setminus D} \to \mathbb{R}$,

$$\mathrm{Cov}(\mathbf{1}_{E_D}, g(X_{V\setminus D})) \leq 0. \tag{6.4}$$

Moreover, writing $b_n = \mathbb{E}S_n$, we have

$$\mathrm{Var}(S_n) < b_n\left(1 - \frac{b_n}{n}\right), \qquad \mathrm{Cov}(X_x, X_y) < 0 \quad (x \neq y). \tag{6.5}$$

At fixed p, a, the densities strictly decrease with dimension:

$$\frac{b_{n-1}}{n-1} > \frac{b_n}{n} \qquad (n \geq 2). \tag{6.6}$$

Here τ_D is the stationary law of the *traced walk*, not the marginal of $\mu_n^{p,a}$. The rate convention in this statement matters.

Lemma 6.2 (Holding, loopless, and trace parameters)**.** *The loopless weighted complete graph with probabilities*

$$\widehat{K}(x,y) = \frac{1}{n-1+a} \quad (y \neq x), \qquad \widehat{K}(x,\partial) = \frac{a}{n-1+a}$$

and sleep rate $\ell > 0$ has the same stable law as (6.1) *at*

$$\lambda = \ell\frac{n+a-1}{n+a}, \qquad p = \frac{\ell(n+a-1)}{n+a+\ell(n+a-1)}. \tag{6.7}$$

For $1 \leq m < n$, tracing $k = n - m$ sites out of the holding model preserves λ, p, a: its kernel is $K^{\mathrm{tr}}(x,y) = 1/(m+a)$ and its killing probability is $a/(m+a)$. In loopless units, the corresponding sleep rate is

$$\ell_{\mathrm{tr}} = \ell\frac{(n+a-1)(m+a)}{(n+a)(m+a-1)}. \tag{6.8}$$

In particular, ordinary simple random walk on K_{n+1} corresponds to $a = 1$, and

$$p = \frac{n\ell}{n+1+n\ell}, \qquad \ell_{\mathrm{tr}} = \ell\frac{n(m+1)}{m(n+1)}.$$

Proof. Erasing a holding probability h changes a sleep rate λ to $\lambda/(1-h)$ when the effective jump rate is normalized to one. This proves (6.7). From any deleted site, the first exit from the deleted block has probabilities proportional to 1 at each surviving site and a at the sink. Summing these excursions gives the claimed holding trace. For clarity, the trace of the original *loopless* kernel has probabilities

$$\widehat{K}^{\mathrm{tr}}(x,x) = \frac{k}{(n+a-1)(m+a)},$$
$$\widehat{K}^{\mathrm{tr}}(x,y) = \frac{n+a}{(n+a-1)(m+a)} \quad (y \neq x),$$

and killing probability $a(n+a)/[(n+a-1)(m+a)]$. Erasing its holding probability gives (6.8). □

6.1. **A three-state lift and its reflection formula.** The auxiliary process has colours $A, B, 0$. Its A sites are called active and its B sites buffered; these are auxiliary colours, not the active and sleeping states during an ARW stabilization. Let I, R, H count these colours, respectively. The occupied set is $O = A \cup B$, with count $W = I + R$. The transitions are as follows:

transition	rate
$A_x \longrightarrow B_x$	q at each A site
$B_x \longrightarrow A_x$	p at each B site
$0_x \longrightarrow A_x \quad (R=0)$	p at each empty site
$(0_x, B_y) \longrightarrow (A_x, 0_y) \quad (R>0)$	p/R for each such pair
$B_y \longrightarrow 0_y \quad (R>0)$	a/R at each B site.

Thus service has total rate a, and each donor choice is uniform among the buffers. Null refreshes may be added so that every site has a rate-one clock with an independent Bernoulli(p) refresh coin.

Lemma 6.3 (Stationary identification and reflection)**.** *The colour chain is irreducible. Its unique stationary occupied law is $\mu_n^{p,a}$. Its active indicators have the autonomous generator of independent Bernoulli refreshes, so their stationary law is the product Bernoulli(p) law. The count generator, on $\Omega_n = \{(i,r) : i, r \geq 0,\ i + r \leq n\}$, is*

$$\begin{aligned} G_n f(i,r) = {} & p(n-i)\big[f(i+1,(r-1)_+) - f(i,r)\big] \\ & + qi\big[f(i-1,r+1) - f(i,r)\big] \\ & + a\mathbf{1}_{\{r>0\}}\big[f(i,r-1) - f(i,r)\big], \end{aligned} \tag{6.9}$$

where terms of zero rate are omitted. Given an initial active field with count i and initial occupied count $s \geq i$, couple its refresh-count path Y with an independent rate-a Poisson process N. Then

$$W_t = \max\left\{ s - N_t,\ \sup_{0 \leq u \leq t} \big[Y_u - (N_t - N_u)\big] \right\}. \tag{6.10}$$

In stationarity the joint law of (I, W) also equals the law of

$$\left(Y_0,\ \sup_{t \geq 0} (Y_t - N_t) \right), \qquad Y_0 \sim \mathrm{Bin}(n,p), \tag{6.11}$$

with fresh future clocks independent of the initial field.

Sampler and reflection mechanism. Each inactive coordinate activates at rate p and each active coordinate deactivates at rate q. This gives the count generator; its three transitions directly verify (6.10). All states communicate through the empty state, so there is a unique stationary law.

The ARW connection uses Junge–Kaufman–Meisel's reservoir sampler [3]. Route every jump through a holding reservoir, and stabilize ordinary sites before releasing its next

particle. An internal release chooses a uniform site, sends any surplus back to the reservoir, and refreshes its remaining particle's state by a fresh Bernoulli(p) coin. A sink release decreases the remaining particle budget. Poissonizing gives rate-one refresh clocks and a rate-a descending boundary. Stabilization occurs when the refresh count meets that boundary.

The stationary reflection can be constructed from the infinite past. Reversibility of the independent refresh field turns its backward maximum into (6.11). The descending-boundary terminal rank has the same maximum distribution: summing the killed birth–death resolvent against the binomial reversible weights identifies its terminal flux with the maximum's mass function. This finite resolvent identity is verified in the companion complete-graph maximum representation, following the JKM sampler. Exchangeability then identifies the full occupied law from its rank law. □

6.2. **The positive averaged source.** Write $\pi_n(i,r)$ for the stationary count law of (6.9). The following comparison concerns the auxiliary active indicators $Z_x = \mathbf{1}_{\{x \text{ has colour } A\}}$.

Lemma 6.4 (An active tag suppresses outside occupancy)**.** *For the colour chain started empty, every $t > 0$, and every tag x,*

$$\mathrm{Law}(O_{-x}(t) \mid Z_x(t) = 1) \preceq_{\mathrm{st}} \mathrm{Law}(O_{-x}(t) \mid Z_x(t) = 0). \tag{6.12}$$

The same comparison holds after prescribing any complete pattern of the other terminal active bits, and in the stationary limit.

Proof. First suppose that the tag has refreshed by time t. Conditional on its clock times, forcing its final active bit changes only its last refresh coin. Couple all earlier randomness. After that last refresh, the two copies have tag colours $(A,0)$ or (A,B) and identical outside colours, or tag colours $(A,0)$ with one outside site buffered only in the second copy. These three forms imply the desired outside inclusion.

They persist under common outside refreshes and service. To see the donor coupling explicitly, for buffer sets C and $C\cup\{z\}$ choose the same uniform element of C with probability $|C|/(|C|+1)$; otherwise choose z in the larger set and a uniform element of C in the smaller. If C is empty, only the larger set chooses a donor. Choosing the special buffer either moves the single outside discrepancy to a common donor or removes it. Refreshing its site has the same effect. All remaining transitions agree. Throughout, the two outside active fields are identical.

There is also a branch in which the inactive tag has never refreshed. It is then permanently empty, and its outside process is the $(n-1)$-site chain from empty. Couple the outside active paths and the service clock with the copy conditioned to have active tag at time t. The extra tag adds between zero and one to Y at every time, hence between zero and one to the reflected total W. Its terminal tag is occupied, so its outside occupied count is no larger. Both outside laws are exchangeable, and a common uniform ordering lifts the count comparison to sets.

This reasoning survives conditioning on the other terminal active bits: their active paths remain identical, prescribed active sites are occupied in both copies, and buffer sets are uniform among the prescribed inactive sites conditional on their counts. Mixing the two tag-clock branches proves the lemma. Finite-state convergence gives the stationary assertion. □

Lemma 6.5 (Nonnegative averaged coefficients)**.** *For $1 \le s \le n-1$ and $0 \le j < s$, set*

$$\kappa_{s,j} = p(n-s)\pi_n(j, s-j) - a\sum_{u=0}^{j} \pi_n(u, s+1-u). \tag{6.13}$$

Then $\kappa_{s,j} \ge 0$. Set $\kappa_{s,-1} = \kappa_{s,s} = 0$.

Proof. Put $\theta_j(r) = \mathbb{P}(R = r \mid I = j)$ and $F_j(r) = \mathbb{P}(R \le r \mid I = j)$, retaining $w_j = \mathbb{P}(I = j)$. Condition on a fixed active pattern of size j and an inactive tag. By exchangeability that tag

is a uniform one of the $n-j$ inactive sites. Lemma 6.4, with the other active bits prescribed, therefore gives

$$F_{j+1}(r-1) \geq F_j(r-1) + \frac{r}{n-j}\theta_j(r). \tag{6.14}$$

Indeed, deleting the inactive tag lowers a buffer count r by one with probability $r/(n-j)$.

Balance across the rectangle $\{I \leq j,\ I+R \leq s\}$, with $r = s-j$, and the binomial identity $p(n-j)w_j = q(j+1)w_{j+1}$ give

$$\kappa_{s,j} = p(n-j)w_j \left[F_{j+1}(r-1) - F_j(r-1) - \frac{r}{n-j}\theta_j(r)\right]. \tag{6.15}$$

For the rectangle, active births cross its right edge, active deaths enter that edge, and service enters through occupied level $s+1$; these are exactly the terms in this identity. Equation (6.14) proves the sign. The endpoint $\kappa_{s,s} = 0$ is balance across the full occupied level $\{W \leq s\}$. □

These coefficients retain the actual stationary averaging. In particular, their consecutive differences need not be nonnegative.

Lemma 6.6 (Technical deletion-source identities)**.** *Write $e_{i,r}$ for unit mass at (i,r) and let G_m^* act on measures. For $1 \leq s \leq m$ and $0 \leq j < s$, define signed measures*

$$B_{s,j} = e_{j,s-j} - e_{j+1,s-j-1}, \qquad D_{s,j} = e_{j,s-j} - e_{j,s-j-1}.$$

For $1 \leq m < n$ and $k = n-m$, put $(h)_k = h(h-1)\cdots(h-k+1)$, $(h)_0 = 1$, and $H_k = \mathbb{E}(n-I-R)_k > 0$. Conditional on a specified k-site block being empty, the remaining coloured count law is

$$\eta_k(i,r) = \frac{(n-i-r)_k \pi_n(i,r)}{H_k}, \qquad i+r \leq m.$$

Its source under the m-site generator at the same p, a is

$$G_m^* \eta_k = \frac{1}{H_k} \sum_{s=1}^{m} k(n-s-1)_{k-1} \sum_{j=0}^{s-1} \kappa_{s,j} B_{s,j}. \tag{6.16}$$

For unconditioned deletion of one site, $m = n-1$, the outside count law is

$$\nu(i,r) = \frac{(n-i-r)\pi_n(i,r) + (i+1)\pi_n(i+1,r) + (r+1)\pi_n(i,r+1)}{n},$$

and its source is

$$G_m^* \nu = \frac{1}{n} \sum_{s=1}^{m} \sum_{j=0}^{s-1} (\kappa_{s,j} - \kappa_{s,j-1}) D_{s,j}. \tag{6.17}$$

Proof outline; coordinate verification in the companion record. The formulas for η_k and ν are sampling without replacement from the three colour classes. Apply full stationarity and compare the full generator on outside functions with G_m. For one deleted tag, an active tag makes no difference. An empty tag may remove an outside buffer when it activates; a buffered tag can shield outside buffers from service or supply a donor to an outside activation. The three discrepancies combine into $\kappa_{s,j} - \kappa_{s,j-1}$, giving (6.17). Weighting by the number $(H)_k$ of ordered vacant tags instead gives (6.16).

This is a finite-generator identity, including the zero-buffer and top-level boundaries; no sign is assigned to the individual discrepancies. The full coordinate and boundary verification is in the companion deletion-source proof, equations (55src) and (55del), with the service-rate modification in "Arbitrary homogeneous sink ratios." The companion `trace_source_check.py` and `trace_source_checks.json` provide finite exact consistency checks for this count model and its source identities. □

6.3. **From the source to the trace sandwich.**

Lemma 6.7 (The two semigroup signs)**.** *Let P_t be the m-site count semigroup, and view an increasing function f of occupied count as $f(i,r) = f(i+r)$. Then*

$$\langle B_{s,j}, P_t f\rangle \le 0.$$

Furthermore, $d_{s,j}(t) = \langle D_{s,j}, P_t f\rangle$ is nonnegative and nonincreasing in j, for $0 \le j < s$.

Proof. Changing one initial buffer to active, at fixed initial occupied count, increases the active path under common refreshes. Reflection (6.10) then increases every later occupied count. This proves the first sign.

For the second, use the same active path Y^j and service clock for initial occupied counts s and $s-1$. With

$$A_j(t) = \sup_{0\le u\le t}[Y^j_u - (N_t - N_u)],$$

the integer-valued reflection gives

$$d_{s,j}(t) = \mathbb{E}\left[(f(s-N_t) - f(s-1-N_t))\mathbf{1}_{\{A_j(t)\le s-1-N_t\}}\right]. \tag{6.18}$$

The integrand is defined as zero when $s-1-N_t < 0$, when its event is impossible. Its function increment is nonnegative and depends only on N_t. Nested initial active sets give $A_j(t) \le A_{j+1}(t)$ under common refreshes. Thus the indicator decreases with j, proving both claims. The adjacent comparison only uses $j \le s-2$, where both lower initial occupied states exist. □

Proof of the trace and block statements in Theorem 6.1. Pair (6.16) with $P_t f$. Its nonnegative coefficients multiply the first, nonpositive semigroup sign. Hence

$$\frac{d}{dt}\langle \eta_k, P_t f\rangle = \langle G_m^* \eta_k, P_t f\rangle \le 0.$$

As $t\to\infty$, the count chain converges to its unique stationary law. Thus the stationary trace count is dominated by the outside count conditioned on the deleted block being empty.

For unconditioned one-site deletion, summation by parts gives

$$n\frac{d}{dt}\langle \nu, P_t f\rangle = \sum_{s=1}^{m}\Bigg[\sum_{j=0}^{s-2}\kappa_{s,j}\big(d_{s,j}(t) - d_{s,j+1}(t)\big) + \kappa_{s,s-1}d_{s,s-1}(t)\Bigg] \ge 0.$$

The adjacent differences, not the original signed source terms, have the required sign. Taking the stationary limit proves that the actual outside marginal lies below its trace. Project this comparison and iterate deletions to obtain it for any block D; tracing preserves p, a and is associative.

All these occupied laws are exchangeable on the surviving sites. A uniform s-subset can be coupled inside a uniform $(s+1)$-subset using one random ordering. Consequently, ordering their counts orders the configurations for every increasing test, without requiring the test to be symmetric. This proves (6.2).

For such a test g, let u, t, b_0, b_1 be its expectations under the outside marginal, the trace, and the conditional laws on E_D^c, E_D, respectively. The comparison just proved gives $u \le t \le b_0$, while

$$u = \mathbb{P}(E_D^c)b_0 + \mathbb{P}(E_D)b_1.$$

Both event probabilities are positive by irreducibility. The mixture identity implies $b_1 \le t \le b_0$, proving (6.3). Finally,

$$\mathrm{Cov}(\mathbf{1}_{E_D}, g(X_{V\setminus D})) = \mathbb{P}(E_D)\mathbb{P}(E_D^c)(b_1 - b_0) \le 0.$$

□

6.4. **Why pair correlation is strictly negative.** Strictness requires a source coefficient which is strictly positive. The following representation supplies one and explains the averaging in the argument.

Lemma 6.8 (Strict source for the mean)**.** *For the one-site deletion law ν,*

$$\frac{d}{dt}\mathbb{E}_\nu W_t\Big|_{t=0} = \frac{1}{n}\sum_{s=1}^{n-1}\kappa_{s,s-1} > 0. \tag{6.19}$$

Strictness mechanism; exact identity deferred. For the test $f(W) = W$, the adjacent buffer-influence differences at time zero vanish, leaving the displayed sum. Its strict positivity is proved by the companion strict-source identity (55strict). In the maximum representation, a future clock environment defines a matroid rank function on initial vacancies. Bernoulli integration rewrites the sum as $p\mathbb{E}[K\mathbf{1}_{\{R=1\}}]$, where K is the matroid rank minus its number of coloops and hence is nonnegative. If the first superposed event is service and the next n events are successful refreshes at distinct sites, the rank function is $B \mapsto \min(|B|, 1)$. With exactly two initial vacancies this gives $R = K = 1$. Both events have positive probability, proving the strict sign. The construction of the rank function and the integration identity are the deferred verification; the remaining variance argument is given below. □

Completion of Theorem 6.1. The mean under one-site deletion starts at $(n-1)b_n/n$, has nonnegative derivative at all times, and has strictly positive initial derivative by Lemma 6.8. Continuity and convergence to the smaller stationary chain give

$$b_{n-1} > \frac{n-1}{n} b_n.$$

On the other hand, the vacant-tag trace comparison and exchangeability give

$$b_{n-1} \le \mathbb{E}[S_n \mid X_x = 0] = b_n - \frac{\mathrm{Var}(S_n)}{n - b_n}.$$

Here $0 < b_n < n$ by full support. Combining these inequalities yields (6.5), since for distinct x, y

$$\mathrm{Cov}(X_x, X_y) = \frac{\mathrm{Var}(S_n) - b_n(1 - b_n/n)}{n(n-1)}.$$

The first inequality is also exactly (6.6). □

Corollary 6.9 (Single-site tests and pure cylinders)**.** *Every occupation indicator is negatively correlated with every increasing function of the other sites. If $\vartheta = \mathbb{E}X_x$ and $D \subseteq V$, then*

$$\mathbb{P}(X_D = 1) \le \vartheta^{|D|}, \qquad \mathbb{P}(X_D = 0) \le (1-\vartheta)^{|D|}.$$

For disjoint $D, F \subseteq V$,

$$\mathbb{P}(X_D = 0,\ X_F = 0) \le \mathbb{P}(X_D = 0)\mathbb{P}(X_F = 0).$$

Proof. Use (6.4) with a singleton block for the first assertion and iterate with the product of the other occupation indicators. For vacancies, apply the singleton assertion to X_x and the increasing function $-\prod_{y\in D\setminus\{x\}}(1 - X_y)$, rearrange, and iterate. For the last assertion, apply (6.4) to the two block-OR indicators and complement both. □

The all-rate conclusion is stronger than pairwise negative correlation: every block-OR is negatively correlated with every increasing outside function. Full NA would also compare higher block-count thresholds. Those comparisons remain open; a conditional law below the trace is not thereby below the actual marginal, which also lies below that trace.

7. Weak killing and gluing

Write $H_D = \{X \cap D = \varnothing\}$. The two BK orientations in this section refer to sleeper indicators X and vacancy indicators $1 - X$.

7.1. **A complete core with rare killing.** Fix $|V| = n \geq 2$, a stochastic kernel invariant under every permutation,

$$P_{ii} = d, \qquad P_{ij} = w > 0 \quad (i \neq j), \qquad d + (n-1)w = 1,$$

and a common sleep rate $\lambda > 0$. Put $K_e = (1-e)P$ and $\nu_e = \mu_{K_e,\lambda}$.

Lemma 7.1 (Quantitative vacancy gap)**.** *There are $e_1 > 0$ and constants $c_\ell > 0$, $1 \leq \ell < n$, depending only on the fixed core data, such that, for every $0 < e < e_1$ and all disjoint nonempty $A, B \subseteq V$ with $|B| = \ell$,*

$$\nu_e(H_A)\nu_e(H_B) - \nu_e(H_{A\cup B}) \geq c_\ell e^\ell \nu_e(H_A).$$

For every prescribed k-set I, $1 \leq k \leq n$,

$$\nu_e(X = V \setminus I) = e^k \gamma_k + O(e^{k+1}), \qquad \gamma_k > 0.$$

Consequently, for $r, \ell \geq 1$ with $r + \ell \leq n$,

$$\gamma_r \gamma_\ell - \gamma_{r+\ell} \geq c_\ell \gamma_r > 0.$$

Quantitative mechanism. The complete-graph deletion proof gives more than a sign. Retain one strictly positive boundary term in its integrated source and use a prescribed sequence of services with no refreshes. For a block of ℓ sites this yields a trace gap of order α^ℓ, where α is the holding model's service rate. The positive source resolvent bounds its coefficient away from zero. These explicit bounds are the deferred calculation in the companion weak-killing proof, (36wc)–(36we), using (55res) and (55pow).

Iterated unconditioned deletion and the vacant-block trace comparison then give $\mathbb{E}[\mathrm{OR}_B \mid H_A] - \mathbb{E}[\mathrm{OR}_B] \geq c\alpha^\ell$. Multiplication by $\nu_e(H_A)$ gives the displayed cylinder gap. The physical conversion is

$$\alpha_e = \frac{e}{(1-e)w}, \qquad p_e = \frac{\lambda}{\lambda + n(1-e)w + e}.$$

The bounds are uniform while p_e stays in a compact subinterval of $(0,1)$, so they transfer to e. Finite absorption with at most n particles gives the exact-hole expansions; a prescribed kill/sleep sequence gives positive leading coefficients. Divide the cylinder gap by $e^{r+\ell}$ to obtain the strict coefficient inequality. □

Theorem 7.2 (Negative association under weak killing)**.** *For the fixed complete core, one $e_0 > 0$ gives strict NA, and strict BK for every pair of nonconstant increasing events, in both orientations, for all $0 < e < e_0$. The one-site NA/BK assertion is immediate.*

More generally, there are a relative neighborhood $\mathcal{U}$ of $(P, \lambda\mathbf{1}, \mathbf{1})$ and $\varepsilon_0 > 0$ such that the same conclusion holds for

$$\widetilde{K}_e(i,j) = (1 - ea_i)\widetilde{P}(i,j),$$
$$(\widetilde{P}, \boldsymbol{\lambda}, \boldsymbol{a}) \in \mathcal{U}, \quad 0 < e < \varepsilon_0.$$

Here $\widetilde{P}$ is stochastic, all rates and a_i are positive, and ε_0 is chosen so $ea_i < 1$. The neighborhood is chosen before $e \downarrow 0$; it may include nonreversible kernels and unequal rates. No volume-uniform or all-killing-strength conclusion is asserted.

Proof outline. For increasing vacancy events on disjoint supports, let r, ℓ be their minimum witness sizes and a_r, b_ℓ their numbers of minimum witnesses. The leading covariance gap is

$$e^{r+\ell} a_r b_\ell (\gamma_r \gamma_\ell - \gamma_{r+\ell}) > 0.$$

For BK with overlapping supports, count distinct unions of disjoint minimum witness pairs. Choose one decomposition of each union; its product coefficient strictly exceeds the union coefficient by Lemma 7.1. Unchosen product terms are positive. If no minimum pair is disjoint, disjoint occurrence has higher order.

Sleeper BK uses the separate "Complementation for sparse BK families" lemma: a full-support sparse BK family whose exact masses satisfy $\mathbb{P}(Z = S)/\mathbb{P}(Z = T) \to 0$ for $T \subsetneq S$

also has BK complements. Its proof assigns each overlap-defect state to a strictly smaller defect state, whose mass dominates. Here the ratios are $O(e^{|S|-|T|})$; the proof also preserves strictness. Arbitrary BK closure under complementation is not being assumed.

There are finitely many event pairs; superlevel decomposition gives the function version of NA. For robustness, graded finite absorption gives uniformly $\nu_e(X = V \setminus I) = e^{|I|}(c_I + O(e))$ with positive, continuous c_I. The finitely many strict inequalities $c_I c_J > c_{I\cup J}$ persist on one neighborhood. Uniform remainder and exact-mass ratio bounds complete the argument; see the robustness proof (36wn). □

7.2. **The response needed for gluing.** For an entrance distribution η on V, make one addition to $X \sim \nu_e$ and stabilize to Y. Set $L = 1 + |X| - |Y|$, the number lost, and use $0^0 = 1$.

Lemma 7.3 (Configuration loss response)**.** *For the fixed complete core, there is $e_2 > 0$ such that for each fixed $0 < e < e_2$ there is $\delta_e > 0$ satisfying*

$$\frac{\mathrm{Cov}(f(Y), u^L)}{1-u} \geq \delta_e \left[f(V) - f(\varnothing)\right]$$

for every increasing real f, every entrance distribution η, and every $0 \leq u < 1$. The quotient extends continuously to $u = 1$ with the same bound. Neither NA nor conditional ordering of L is a hypothesis or a conclusion of this lemma.

Proof outline. Let $A_v(Y, X)$ be the probability of closed addition at v taking input X to output Y, used only with at most n total particles. For a terminal hole set I of size $1 \leq r < n$, the closed-column calculation (A1)–(A10) proves

$$S_v(I) := \sum_{|J|=r+1} A_v(V \setminus I, V \setminus J) \leq \frac{n}{r+1}.$$

Its mechanism is a positive tagged-site bias under addition to a uniform fixed-rank input. The bias coefficients obey (A6), obtained by commuting closed additions and exposing the forced first jump from an occupied tag. That jump has no extra sleep-failure factor. The full output rank is deterministic and is handled separately.

For a nonconstant increasing indicator f, let r be the smallest number of holes making it fail, $\mathcal{F}_r$ the corresponding hole sets, $C_f = |\mathcal{F}_r|\gamma_r$, and $M = n\gamma_1$. Set $S_\eta(I) = \sum_v \eta_v S_v(I)$. No loss has probability $eM + O(e^2)$; a failing r-hole output with no loss requires an $(r+1)$-hole input. Using $\gamma_{r+1} \leq \gamma_1\gamma_r$ from the cylinder comparison, the no-loss covariance coefficient satisfies

$$d_{f,\eta} = MC_f - \gamma_{r+1} \sum_{I\in\mathcal{F}_r} S_\eta(I) \geq \frac{r}{r+1} MC_f > 0.$$

For $r = n$ the subtracted term is zero. This is a configuration calculation, not a symmetrization of an arbitrarily driven addition.

Proper-input transition columns are analytic at $e = 0$. Stationarity reconstructs the full-input column via (A15), avoiding any claim of closed stabilization of $n + 1$ particles. A terminal r-hole failure comes at leading order from full input, hence has $L = r + 1$; the unconditioned limiting loss is one. Since $L \leq n + 1$, polynomial division by $1 - u$ and the two endpoint expansions give

$$\begin{aligned}\frac{\mathrm{Cov}(f(Y), u^L)}{1-u} &= e^r C_f u(1 + \cdots + u^{r-1}) \\ &\quad + e^{r+1}\{d_{f,\eta} + u a_{f,\eta}(u)\} + O_{\mathrm{coeff}}(e^{r+2}).\end{aligned}$$

The polynomial coefficients and remainders are uniformly bounded over all entrances: mixtures are linear in the finitely many pure entrances. For sufficiently small e, the quotient is bounded below by

$$\tfrac{1}{2} C_f e^r u + \frac{r}{2(r+1)} MC_f e^{r+1} > 0,$$

including both endpoints. Finitely many indicators and superlevel decomposition yield the stated uniform margin. Full algebra and boundary checks are in the configuration-response proof, (A11)–(A18). □

Proposition 7.4 (A weighted-column criterion)**.** *For any fixed irreducible stochastic core P and positive site rates, use $K_e = (1-e)P$, write its exact-hole coefficients as $c_I > 0$, and put $M = \sum_i c_{\{i\}}$. The same uniform loss-response conclusion holds if, for every entrance vertex v and nonempty I,*

$$Mc_I > \sum_{|J|=|I|+1} c_J A_v(V \setminus I, V \setminus J),$$

with the empty sum zero. Indeed, summing these weighted gaps over $\mathcal{F}_r$ and averaging in η gives the preceding $d_{f,\eta}$. This criterion is unproved in general and does not supply isolated-core NA. Positivity of the kernels alone supplies neither this inequality, preservation of constants, nor reversibility.

7.3. **A complete core with product recipients.** A coordinate is essential to a function if changing that coordinate can change its value.

Theorem 7.5 (Gluing a complete core to independent sites)**.** *Attach $m \geq 1$ recipient sites W with fixed rates $\mu_i > 0$. For $t_i \geq 0$, $0 < \sum_i t_i \leq 1$, and an arbitrary probability vector η on V, set*

$$K_{VV} = (1-e)P, \qquad K_{Vi} = et_i \mathbf{1}_V,$$
$$K_{iV} = b\eta^T, \qquad K_{ij} = 0 \quad (i,j \in W).$$

Remaining row mass goes to the sink. Thus all recipients have the same *return probability and entrance law, and forward routing fractions are independent of the core exit vertex.*

There is $e_* > 0$ such that, for each fixed $0 < e < e_*$, one $b_0(e) \in (0,1)$ gives full site NA and BK in both orientations for every permitted $\boldsymbol{t}, \eta$ and $0 < b < b_0(e)$. For disjoint nonconstant increasing functions, covariance is strictly negative exactly when both essential supports meet $V \cup \{i : t_i > 0\}$. Zero-weight recipients are independent upstream singleton components. Thresholds depend on the fixed core and recipient data; no uniform b-interval as $e \downarrow 0$, separate recipient entrances, or unequal return probabilities are asserted.

Proof outline. At $b = 0$ the law is $\nu_e \otimes \pi_W$, where $\pi_W = \bigotimes_i \mathrm{Ber}(p_i)$ and $p_i = \mu_i/(1+\mu_i)$. For F using core coordinates and recipients S, and a recipient-only G on disjoint coordinates T, let $\bar{F}$ average over S under π_W, and put $c_i(G) = \mathrm{Cov}_{\pi_W}(X_i, G)$. Marking one returned departure and retaining the changing means gives

$$[b]\,\mathrm{Cov}(F,G) = -\sum_{i \in T} c_i(G)\,\mathrm{Cov}(\bar{F}(Y), (1-t_i)^L)$$
$$\leq -\delta_e \,\mathrm{osc}(\bar{F}) \sum_{i \in T} t_i c_i(G).$$

Here $[b^k]$ denotes a Taylor coefficient and $\mathrm{osc}(\bar{F}) = \bar{F}(V) - \bar{F}(\varnothing)$. The factor $(1-t_i)^L$ is the probability that no core loss refreshes recipient i. The exact signed-measure identity is (67first); the bound is (A21).

For two recipient-only tests on disjoint supports S, T, the first coefficient vanishes and the two-return calculation (67two), (67leaf) gives

$$[b^2]\,\mathrm{Cov}(F,G) = -\sum_{i \in S, j \in T} t_i t_j H_e(t_i, t_j) c_i(F) c_j(G).$$

This coefficient uses the *ordered successive* loss PGF $G_{2,e}(z,w) = \mathbb{E}[z^{L_1} w^{L_2}]$: these are losses in two successive stationary additions with the same entrance law η and fresh instruction

randomness, not independent losses. With $G_e(z) = \mathbb{E}[z^L]$ and $s = 1 - t - u$,

$$\begin{aligned} tuH_e(t,u) = {} & G_e(1-t)G_e(1-u) + G_{2,e}(s,s) \\ & - G_{2,e}(1-t,s) - G_{2,e}(1-u,s). \end{aligned}$$

Stationarity makes the numerator divisible by tu. At $e = 0$ the terminal-kernel continuation makes a full core emit exactly one particle, so $G_0(z) = z$, $G_{2,0}(z,w) = zw$, and $H_0 = 1$. Consequently $H_e \geq 1/2$ for small e, uniformly over $t, u \geq 0$, $t + u \leq 1$, and η.

The queued-block sampler has finitely many states: occupied particles plus queued arrivals never exceed $n + m$. At $e = 0$ a proper core absorbs an arrival and a full core routes one particle. Process core queues, then prescribe successful recipient sleeps and kill their emitted particles. This empties every queue in at most $2(n + m)$ steps with uniformly positive probability for $b \leq b_* < 1$. Its finite absorption resolvent therefore gives uniform analyticity near zero, including total forward routing one and all entrance faces; see (67ext)–(67router) and the uniformity checks after (A21). If both tests use the core, strict core NA controls the constant coefficient. Otherwise exact independence on zero-routing faces factors the covariance as brJ, or b^2rsJ for two recipient-only tests, where r, s are the total routing weights on their essential recipient supports. The displayed response formulas make the normalized leading coefficients uniformly negative. Parametrizing routing within each support by a probability vector handles the zero faces. Compactness and finitely many indicator pairs give one $b_0(e)$; superlevel decomposition completes NA.

BK requires a separate step. The strict-core/product neighborhood lemma (41bk) says that every NA law sufficiently close to a full-support strict-BK core times a Bernoulli product is BK. Overlapping essential supports have a positive reference BK gap; disjoint supports use NA. Apply it in both orientations, using Theorem 7.2 and uniform convergence as $b \downarrow 0$. The loss-response and two-return lemmas themselves do not assume NA; the isolated-core NA and strict BK inputs enter only at these final steps. □

8. Consequences for fluctuations, density, and rank

The NA conjecture has quantitative consequences even without an explicit stationary distribution. In this section write $\theta_v = \mathbb{E}X_v$ for the *actual stationary occupation probability*, rather than the probability of a sleep instruction. The next proposition applies to every NA Bernoulli vector, and therefore unconditionally in each regime where NA was proved above.

Proposition 8.1 (Product moments and concentration)**.** *Suppose X is NA. For $A \subseteq V$,*

$$\mathbb{P}(X_A = \mathbf{1}) \leq \prod_{v \in A} \theta_v, \qquad \mathbb{P}(X_A = \mathbf{0}) \leq \prod_{v \in A} (1 - \theta_v). \tag{8.1}$$

For nonnegative weights w_v, put $S = \sum_v w_v X_v$, $m = \mathbb{E}S$, and $Q = \sum_v w_v^2$. Then, for every $t \in \mathbb{R}$,

$$\mathbb{E}e^{t(S-m)} \leq \prod_v e^{-tw_v\theta_v}(1 - \theta_v + \theta_v e^{tw_v}). \tag{8.2}$$

In particular,

$$\operatorname{Var}(S) \leq \sum_v w_v^2 \theta_v(1 - \theta_v) \leq Q/4, \tag{8.3}$$

and, for $Q > 0$ and $r > 0$,

$$\mathbb{P}(|S - m| \geq r) \leq 2\exp(-2r^2/Q). \tag{8.4}$$

Proof. Two decreasing functions also have nonpositive covariance, by negating both. Induction therefore gives $\mathbb{E}\prod_j h_j \leq \prod_j \mathbb{E}h_j$ for nonnegative functions on disjoint supports that are all increasing or all decreasing. Apply this to X_v, $1 - X_v$, and $e^{tw_v X_v}$ to obtain (8.1)–(8.2). Expanding the variance and using pairwise NC proves (8.3).

For $0 \le \theta \le 1$, let $\psi_\theta(s) = \log(1-\theta+\theta e^s) - \theta s$. We have $\psi_\theta(0) = \psi'_\theta(0) = 0$ and $\psi''_\theta(s) \le 1/4$, so $\psi_\theta(s) \le s^2/8$ for both signs of s. Thus (8.2) is at most $e^{t^2Q/8}$. Exponential Markov inequality with $t = 4r/Q$ gives the upper tail $e^{-2r^2/Q}$; using $-t$ gives the lower tail. □

The first bound gives a simple hole estimate for every region:

$$\mathbb{P}(N_A = 0) \le \exp(-\mathbb{E}N_A). \tag{8.5}$$

For an unweighted count of mean $m > 0$, the same product moment proof gives the mean-sensitive upper tail

$$\mathbb{P}(N \ge (1+\delta)m) \le \exp\{-m[(1+\delta)\log(1+\delta) - \delta]\}, \qquad \delta > 0.$$

Indeed $\log \mathbb{E}e^{t(N-m)} \le m(e^t - 1 - t)$; optimize at $t = \log(1+\delta)$. This is useful when the mean is much smaller than the volume.

Corollary 8.2 (All-rate complete-graph concentration)**.** *All bounds* (8.1)–(8.5) *hold on complete graphs at every size, common positive rate, and homogeneous positive sink weight, even though full NA there remains open.*

Proof. Theorem 6.1 with a singleton block gives $\mathrm{Cov}(X_v, g(X_{-v})) \le 0$ for every increasing g. For $t \ge 0$, write $e^{tw_vX_v} = 1 + (e^{tw_v} - 1)X_v$ and apply this comparison to the product of the other exponential factors. For $t < 0$ that product is decreasing, so its covariance with X_v is nonnegative; the coefficient $e^{tw_v} - 1$ is nonpositive and reverses the inequality. Induct to obtain (8.2); the cylinder and variance arguments use the same singleton comparison. Proposition 8.1's scalar estimates then follow unchanged. □

Corollary 8.3 (From a mean density to a concentrated density)**.** *Suppose stationary laws on V_k are NA, $n_k = |V_k| \to \infty$, and $\mathbb{E}N_{V_k}/n_k \to \rho$. Then $N_{V_k}/n_k \to \rho$ in mean square and probability. For every $\varepsilon > 0$, eventually*

$$\mathbb{P}(|N_{V_k}/n_k - \rho| \ge \varepsilon) \le 2e^{-\varepsilon^2 n_k/2}.$$

The same assertion holds under the singleton comparisons used in Corollary 8.2.

Proof. The mean square error is at most $1/(4n_k) + (\mathbb{E}N_{V_k}/n_k - \rho)^2$. Eventually the mean bias is less than $\varepsilon/2$; apply (8.4) with $r = \varepsilon n_k/2$. □

This is a precise conditional link to the ARW density questions. Levine–Silvestri distinguish convergence of the stationary density, identification with the fixed-energy critical density, and existence of a microscopic limit [4]. If a separate theorem identifies the mean limit as ρ_c, Corollary 8.3 concentrates the stationary density at ρ_c. NA alone cannot identify that mean: independent Bernoulli laws are NA at every density. In dimension one, Hoffman–Johnson–Junge already prove the relevant density equalities and exponential stationary-count concentration without assuming NA [5, Theorem 1.1, Propositions 8.5–8.6].

Proposition 8.4 (Inheritance by a limiting field)**.** *If finite-volume NA laws have convergent finite-coordinate marginals, the limiting field is NA for increasing cylinder functions on disjoint finite supports. Increasing functions of disjoint coordinate blocks, including regional particle counts, also form an NA family.*

Proof. For fixed finite supports, the three expectations in (1.1) are finite sums of marginal probabilities, so the inequality passes to the limit. For the block assertion, compose the two increasing tests with the increasing block functions; their original supports remain disjoint. □

This does not supply existence or uniqueness of the microscopic limit. In particular, graph-dependent perturbative thresholds do not yield NA at one fixed λ along an arbitrary exhaustion.

8.1. **Hyperuniformity requires quantitative cancellation.** Hyperuniformity along growing regions means $\mathrm{Var}(N_A) = o(|A|)$; the lattice conjecture in [4, Section 8.1] asks for a stronger subvolume power bound under its proposed limiting law. The identity

$$\mathrm{Var}(N_A) = \sum_{v\in A} \theta_v(1-\theta_v) + 2 \sum_{\{u,v\}\subseteq A} \mathrm{Cov}(X_u, X_v)$$

shows the distinction: NA supplies signs, while hyperuniformity requires the negative covariance sum to cancel the diagonal sum to leading order. This implication cannot hold for arbitrary sink graphs. A sink joined to n otherwise isolated vertices has the product stationary law $\mathrm{Bernoulli}(\lambda/(1+\lambda))^{\otimes n}$, by Proposition 2.2. It is NA and has variance proportional to n.

On complete graphs there is a quantitative positive result beyond this sign information. Put $p = \lambda/(1+\lambda)$ for fixed loopless sleep rate $\lambda > 0$.

Theorem 8.5 (Complete-graph fluctuation consequence of JKM)**.** *For ordinary loopless K_{n+1}, the stationary count S_n satisfies*

$$\frac{\mathbb{E}S_n}{n} \longrightarrow p, \qquad \mathrm{Var}(S_n) \sim \frac{\pi^2 p(1-p)}{6}\frac{n}{\log n}, \qquad \mathrm{Cov}(X_i, X_j) \sim -\frac{p(1-p)}{n} \quad (i \neq j).$$

Published asymptotic and parameter transfer. Junge–Kaufman–Meisel prove the mean and variance asymptotics for the holding complete graph at fixed refresh probability p [3]. The loopless model corresponds to $p_n = n\lambda/(n+1+n\lambda)$, with $n|p_n - p| \le 1/4$. Couple the two maximum representations using the same clocks and uniform refresh coins. At each refresh epoch the discrepancy count is $\mathrm{Bin}(n, |p_n - p|)$. Both maxima occur before the nth service; there are n^2 expected refreshes before that time. A union bound and the binomial tail therefore give $\|S_n(p_n) - S_n(p)\|_2 = O(\log n)$. The full estimate is in the companion parameter-coupling proof following (55NC). Since $\log n = o(\sqrt{n/\log n})$, this difference transfers the variance asymptotic as well as the mean limit. Finally exchangeability gives

$$\mathrm{Cov}(X_i, X_j) = \frac{\mathrm{Var}(S_n) - n\theta_n(1-\theta_n)}{n(n-1)}, \qquad \theta_n = \mathbb{E}S_n/n \to p,$$

which proves the pair asymptotic. □

The variance scale is an input from JKM, with the moving-parameter issue handled by the coupling. It is stronger than the linear variance bound from negative dependence alone.

8.2. **Rank log-concavity as a sufficient route.** For an exchangeable law on n sites, let $r_k = \mathbb{P}(N = k)$ and $a_k = r_k/\binom{n}{k}$. Ultra log-concavity (ULC) means that (a_k) is log-concave with interval support.

Proposition 8.6 (An exchangeable sufficient condition)**.** *An exchangeable ULC law is NA. In particular, an all-rate ULC theorem for the complete-graph stationary rank law would settle its full NA conjecture.*

Proof. Conditional on a pattern with j occupied coordinates, a remaining set of size ℓ has weight proportional to $a_{j+\ell}$. Log-concavity makes $a_{j+1+\ell}/a_{j+\ell}$ decrease with ℓ, so increasing the conditioned count stochastically decreases the remaining count. Nested uniform subsets lift this to increasing configuration tests. For disjoint increasing f, g, average f over its block conditional on that block's count. This average is increasing, whereas the conditional expectation of g is decreasing. Their covariance is nonpositive. Interval-support endpoints are handled by the same monotone support ordering. See also [6, Theorem 3.7]. □

The converse fails even for exchangeable laws. For three sites the polynomial $(3 + 2\sum_i z_i + 2\sum_{i<j} z_i z_j)/15$ defines an NA law: the singleton-versus-singleton, AND, and OR covariances are respectively $-2/75, -4/75, 0$, exhausting the increasing-event tests. Its normalized rank sequence $(3,2,2,0)/15$ is not log-concave [6, Example 2.2]. Thus ULC is a sufficient stronger target. Similarly, BK supplies inequalities for disjoint witnesses and implies NA, but NA alone does not supply BK [7].

9. Finite certificates and sharp limitations

Theorem 9.1 (All rates through four sites)**.** *Every finite undirected simple sink graph with at most four nonsink vertices has an NA stationary law at every common $\lambda > 0$. Ordinary complete graphs are also covered through five nonsink vertices.*

Exact certificate proof. For four sites there are 44 graphs with connected nonsink graph, up to relabeling fixing the sink. The exact change of basis in the proof of Theorem 2.3 gives their sixteen stationary masses as $W_B(t)/D(t)$, with $t = 1/\lambda$ and $D(t) > 0$. For every disjoint increasing-event pair E, F, clear denominators:

$$\mu(E)\mu(F) - \mu(E \cap F) = \frac{(\sum_{B\in E} W_B)(\sum_{B\in F} W_B) - D\sum_{B\in E\cap F} W_B}{D^2}.$$

Each numerator has nonnegative coefficients. The 44 exact laws and certificates and their checker supply all 174 comparisons per graph. Disconnected cases factor, and threshold decomposition gives NA. The five-site complete case follows from the verified rank inequalities and exchangeability; its separate calculation is in the complete-graph rank proof and complete-graph checker. □

This is a proof for all positive rates on the listed graphs, rather than sampling at selected values of λ. It also makes the following separation between dependence properties especially concrete.

Let $P_\mu(z) = \mathbb{E}_\mu \prod_v z_v^{X_v}$. The law is *Rayleigh* when pairwise negative correlation survives every positive external field $\prod_v z_v^{X_v}$. It is *strongly Rayleigh* when P_μ has no zero with all variables in the upper half-plane. Strong Rayleigh implies NA [6].

Proposition 9.2 (NA does not extend to conditional NC or Rayleigh in ARW)**.** *Take internal edges $0--1--2$ and sink edges $0--\partial$ and $1--\partial$, with common sleep rate λ. The stationary law is NA for every $\lambda > 0$, but*

$$\operatorname{Cov}(X_1, X_2 \mid X_0 = 0) = \frac{4}{1215\lambda^2} + O(\lambda^{-3}) > 0 \tag{9.1}$$

for sufficiently large λ. It is therefore neither conditionally negatively correlated nor Rayleigh in that range.

Proof. Theorem 9.1 proves NA. The escape recurrence (4.3) on this three-site graph gives

$$C_{\{0\}} = \frac{3}{4}, \qquad C_{\{0,1\}} = \frac{1}{3}, \qquad C_{\{0,2\}} = \frac{77}{180}, \qquad C_{\{0,1,2\}} = \frac{26}{135}.$$

For vacancies, conditioning on $Y_0 = 1$ and using (4.1) gives the coefficient

$$\frac{C_{\{0\}}C_{\{0,1,2\}} - C_{\{0,1\}}C_{\{0,2\}}}{C_{\{0\}}^2} = \frac{4}{1215}.$$

Complementing the two tested variables preserves covariance, proving (9.1). At each sufficiently large fixed rate, the tilted covariance tends to this positive value as $z_0 \downarrow 0$, with $z_1 = z_2 = 1$. Continuity supplies a strictly positive field violating Rayleigh. An exact finite-rate witness at $\lambda = 64$ and $z_0 = 1/100$ is recorded in the three-site certificate. □

Proposition 9.3 (Stabilization does not preserve NA)**.** *On the two-site graph with both sites adjacent to the sink, set $\lambda = 1$. Starting from the deterministic sleeper set $\{1\}$, make one uniform addition and stabilize. The output masses, in states $00, 10, 01, 11$, are*

$$\frac{1}{156}(25, 19, 58, 54).$$

Consequently $\operatorname{Cov}(X_0, X_1) = 31/3042 > 0$, although the input is deterministic and hence NA.

Proof. The finite first-step equations give the displayed law; both addition kernels and their uniform mixture are recorded in the two-site calculation. Substitution gives $54/156 - (73/156)(112/156) = 31/3042$. □

Thus the negative-dependence results use the stationary sampler and its identities. Neither preservation from arbitrary initial laws nor closure under conditioning can replace those arguments.

10. Precise remaining questions

The general conjecture asks for full NA at intermediate parameters on arbitrary undirected graphs. The results above give several more focused questions with explicit mathematical content.

On complete graphs, the block-OR theorem handles $\{N_A \geq 1\}$ against every increasing outside test. The missing extension is to $\{N_A \geq r\}$ for $r \geq 2$. The companion centered deletion identities, (55tail) and (55diff)–(55comp), express these covariances as averaged source terms. Some individual source contributions are negative; the desired statement is an inequality for their actual stationary average.

For weak killing of a general irreducible reversible kernel, write $\mu_e(X = V \setminus I) = e^{|I|}c_I + O(e^{|I|+1})$. Even the general pair inequality $c_{\{i\}}c_{\{j\}} \geq c_{\{i,j\}}$ remains open. The weighted-column criterion stated above isolates a different sufficient condition for the configuration loss response. Proving that criterion would extend the response theorem; a site-NA gluing theorem would still need NA of the isolated core.

11. Companion proofs and exact checks

The following Git files verify the explicitly stated technical inputs and supply the exact certificates. Finite checks establish only their specified families; the general theorems use the analytic arguments.

(i) Full-occupation identity: constant-term calculation (22); `algebra.py` checks the change of basis.

(ii) Large-rate separator correction: formulas (14)–(17), including zero escape probabilities; `vacancy.py` checks the expansion.

(iii) Weak internal motion: deletion identities (38)–(41); `weak_interaction.py` gives finite checks.

(iv) Complete-graph deletion sources: (55act)–(55NC); quantitative weak-killing bounds (36wc)–(36wn).

(v) Configuration loss and gluing: (A1)–(A21); return expansion (67core)–(67leaf); BK neighborhood (41bk). `weak_killing.py` and `avalanche_response.py` give exact diagnostics.

(vi) Small-graph certificates: all 44 four-site laws, checked by `four_site.py`. The three-site file records the counterexamples.

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